

SCHOOL FEE PAYMENT AND HOSTEL ALLOCATION SYSTEM

Adeleke University, Ede

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Statement of Problem



Time Inefficiency

Students spend hours in queues for payments and hostel allocation



Error-Prone Records

Manual data entry leads to payment discrepancies and lost records



Heavy Workload



Administrative staff overwhelmed with manual verification processes



Security Risks



Manual handling vulnerable to fraud and data breaches

Impact at Adeleke University

- Long queues (2+ hours)
- Lost payment receipts
- Hostel allocation delays
- Lack of transparency
- No real-time tracking
- Manual verification errors

Key Features



Secure Payment



Paystack integration with multiple payment options



Authentication



Role-based access control for students & admins



Admin Dashboard



Real-time payment tracking and management



Auto Allocation



FCFS algorithm for fair hostel room assignment



Auto Receipts



Instant email confirmation with unique references



Responsive Design



Mobile-friendly web interface for all users

Literature Review

Payment Systems Research

Adebayo & Olamide (2021)

Digital Payment Adoption

- ✓ Identified key factors for adoption in Nigerian universities
- ✗ No actual system implementation

Johnson & Adekunle (2023)

Online Fee Payment Portal

- ✓ Secure payment with Flutterwave integration
- ✗ Limited to fees only, no other services

Hostel & Cloud Systems

Okonkwo & Eze (2020)

Automated Hostel Allocation

- ✓ Successfully automated room assignment process
- ✗ No payment integration or verification

Ekwueme (2021)

Cloud-Based Student Systems

- ✓ Robust framework for cloud infrastructure
- ✗ Theoretical only, no practical implementation

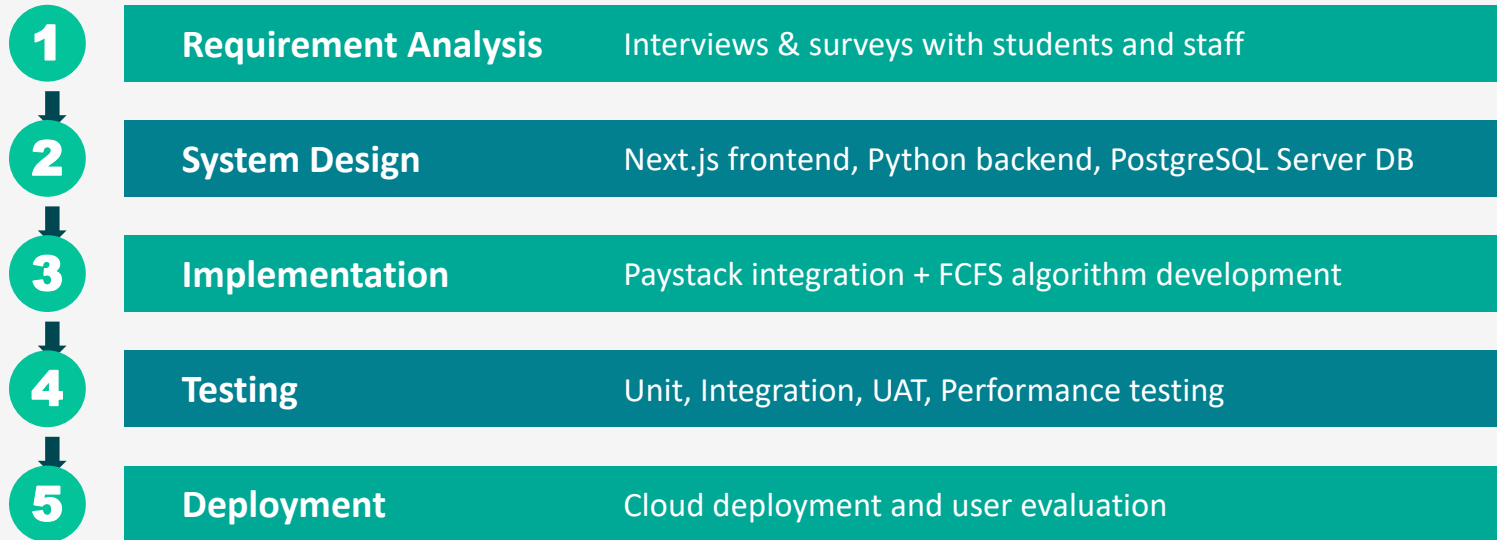


RESEARCH GAP

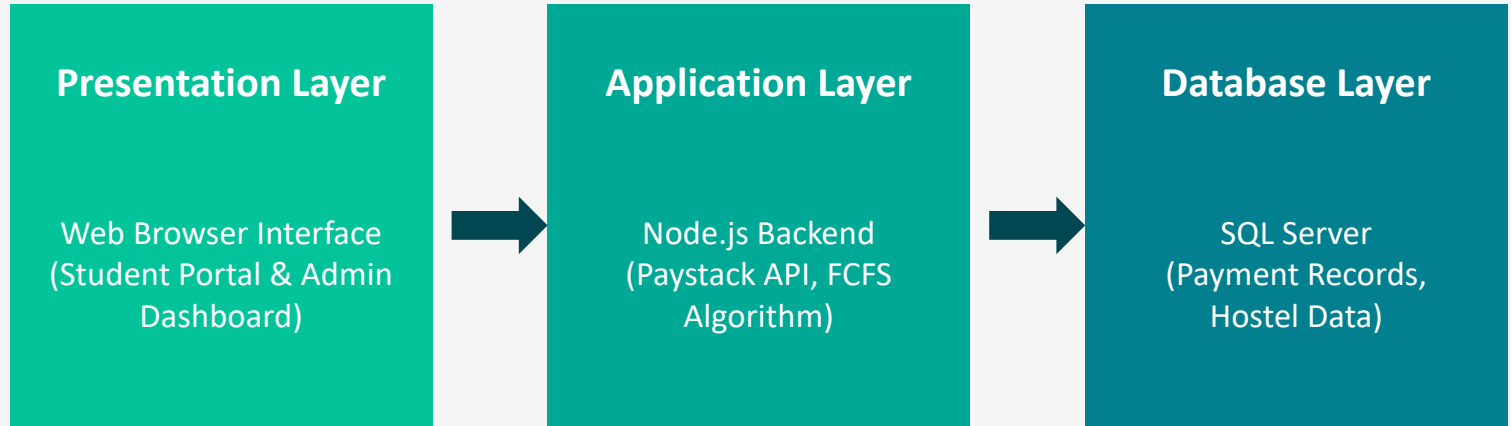
No integrated system exists that combines secure online payment processing with automated hostel allocation specifically designed for Nigerian universities. Current solutions are fragmented, addressing either payment OR allocation, but not both in a unified platform.

Proposed Methodology

Agile Development (Scrum Framework)



System Architecture



Key Technologies: Next.js • Python(FastApi)• PostgreSQL Server • Paystack API • FCFS Algorithm

Significance of the Study

Expected Impact

95%

Reduction in processing time

Zero

Manual payment errors

24/7

System availability

100%

Fair allocation (FCFS)

Key Benefits

- ⚡ Enhanced operational efficiency
- ✓ Improved data accuracy
- ◆? Enhanced security measures
- ◆?/? Better user experience
- ◆?/? Cost-effective cloud solution
- ◆?/? Real-time insights & analytics
- ◆?/? Digital innovation in education
- ◆?

Conclusion

This research proposes an integrated cloud-based system that addresses the inefficiencies in school fee payment and hostel allocation at Adeleke University. By leveraging Paystack for secure payments and FCFS algorithm for fair allocation, the system will enhance transparency, reduce processing time, and improve overall user satisfaction.

Key Takeaways:

- ✓ Automates manual processes, saving time and reducing errors
- ✓ Ensures fair, transparent hostel allocation using FCFS
- ✓ Provides real-time access to payment and allocation data
- ✓ Scalable cloud-based solution for future expansion